

IBM HR Attrition & Retention Analysis

Comprehensive Project Summary Report | Focus: 5-to-6 Year Tenure Windows & Performance Optimization

1. Executive Summary

Employee attrition represents a critical operational challenge for modern enterprises, impacting institutional knowledge, project momentum, and recruitment overhead. This project investigates the benchmark **IBM HR Analytics Employee Attrition Dataset** through exploratory data analysis and custom visualizations, dissecting turnover patterns across tenure brackets with a specific emphasis on the critical 5-to-6-year mid-career threshold.

1,470

TOTAL EMPLOYEES ANALYZED

16.1%

OVERALL ATTRITION RATE

5-6 Yrs

CRITICAL PLATEAU WINDOW

2. Key Numerical Findings & Dataset Metrics

Through systematic data extraction and aggregation via Python (Pandas and Seaborn), the following quantitative performance metrics were established:

Metric Category	Benchmark / Value	Analytical Insight
Total Dataset Records	1,470 Rows, 35 Columns	Comprehensive multi-departmental corporate workforce record.
Total Attrition Volume	237 Employees (16.1%)	Baseline organizational turnover requiring targeted mitigation.
Average Monthly Income	\$6,502.93 overall	Significant gap between retained staff (\$6,832) and leavers (\$4,781).
Tenure Peak Turnover	Years 1 & Years 5-6	Initial onboarding attrition followed by mid-career stagnation drops.
Overtime Correlation	~30.5% Attrition Rate	Staff with frequent overtime exhibit triple the attrition of non-OT staff (~10.4%).

3. Visualization Theme & Methodology

In accordance with project specifications, custom conditional color-mapping was implemented using Matplotlib to visually emphasize directional trends:

- **Yellow Palette (Attrition Rate by Tenure):** Light yellow (#FFF59D) signifies stable or declining turnover rates, whereas dark yellow/amber (#F57F17) highlights upward surges in attrition across specific tenure years.
- **Blue Palette (Monthly Income Progression):** Light blue (#BBDEFB) represents baseline or stagnant earnings, transitioning to deep navy blue (#0D47A1) when average monthly income scales upward with seniority.

Tenure Insight: While Year-1 turnover stems from cultural adjustment and onboarding friction, the secondary turnover wave at Years 5-6 is driven directly by *career plateauing and compensation mismatch*.

4. Root Cause Analysis: Why Employees Leave

Synthesizing statistical correlations and behavioral indicators highlights three primary drivers of voluntary resignation:

- **The 5-to-6 Year Career Ceiling:** Individuals remaining in identical roles for 5 to 6 years without promotions or expanded responsibilities experience professional stagnation.
- **Compensation Stagnation:** Internal salary adjustments failing to keep pace with broader market inflation for mid-level specialists.
- **Burnout & Excessive Workload:** Unbalanced overtime requirements without adequate recovery windows accelerate exhaustion among high-impact talent.

5. Strategic Recommendations & Performance Boost

To curb attrition and maximize overall corporate performance, organizations should adopt the following framework:

Actionable Retention & Performance Plan:

1. **Structured Career Roadmaps:** Introduce mandatory career advancement and lateral mobility reviews at 3-year and 5-year intervals.
2. **Proactive Compensation Benchmarking:** Conduct regular market-rate salary reviews for mid-tenure employees to prevent external poaching.
3. **Workload Balancing & Flexibility:** Enforce strict overtime monitoring and mental health support to safeguard employee stamina and productivity.